

Dark current spikes as an indicator of mobile dislocation dynamics under intense electric fields

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Breakdown of metals subject to intense electric fields is a long-standing problem in high-gradient applications. A stochastic model was proposed, in which breakdown is nucleated by a critical transition in the density of mobile dislocations near the metal surface. Prior to breakdown, the density fluctuates around a stable point, with a finite probability of reaching the critical point at any time. It was suggested that these prebreakdown fluctuations can lead to corresponding dark current fluctuations. We constructed a setup allowing the measurement of such fluctuations, in which prebreakdown current spikes were detected. The dependence of the spike rate on the field, as well as the distribution of times between spikes, were as predicted by the model. This validates the model, and provides a possible early-warning signal of imminent breakdown.

Breakdown of metals subject to intense electric fields is a long-standing problem in high-voltage applications, limiting the obtainable amplitudes of electric and electromagnetic fields.^{1–4} One specific application is the compact linear collider (CLIC) designed in CERN and other high-gradient accelerators, where understanding and controlling breakdown are pivotal topics.⁵

Surface features have been postulated to be essential in the development of breakdown.^{6–8} However, after an initial state of conditioning, breakdown rates normally saturate at an approximately constant level.^{9,10} This suggests that breakdown sites are intrinsic and formed during operation, since preexisting surface formations or extrinsic sources would be depleted over time. In addition, prebreakdown surface features have not been observed experimentally.^{3,6,7,11}

We previously proposed^{12,13} that a critical transition in the density of the mobile dislocations within a metal can lead to a rapid increase in local current. This mobile dislocation density fluctuation (MDDF) model provides a description of the initial stage of breakdown nucleation, showing that intense electric fields can cause surface response, thus nucleating breakdown.^{12,13} This description is in line with previous suggestions that the electric field and current are enhanced due to the formation of a protrusion, leading to localized heating and plasma formation through evaporation,^{11,14} and thus to breakdown.^{1,15,16} However, in the MDDF model, the localized plastic activity leading to this enhancement is the result of a rapid transition in the mobile dislocation population, which does not produce observable features prior to breakdown.

In addition to providing a qualitative explanation of the characteristics of breakdown nucleation, the MDDF model also provides statistical tools to calculate breakdown rates, which are in agreement with experimental

observations.^{12,13,17,18} Although breakdown occurs in the model only when there is a critical transition, it was proposed that subcritical fluctuations of the mobile dislocation density might be observed as fluctuations of the dark current, prior to breakdown, in electrodes exposed to high electric fields.^{12,13} Such measurements have now been carried out at CERN in a dedicated system, and the purpose of this manuscript is to compare their results to the time distribution of fluctuations derived from the MDDF model.

Measurements of prebreakdown dark currents were performed in the Large Electrode System (LES) at CERN. The LES is designed to test various electrodes under high-gradient conditions, enabling application of electric fields up to ~ 100 MV/m (see Refs. 19 and 20 for a detailed description). Here we used oxygen-free high conductivity (OFHC) Cu electrodes, featuring a machined $10 \text{ mm} \times 1 \text{ mm}$ rectangular ridge, and separated by a gap of $60 \text{ }\mu\text{m}$. These ridges were initially arranged crosswise, giving about 1 mm^2 area exposed to the high field, and then rotated to be parallel, increasing the area to 10 mm^2 (see Fig. S1). The conditioning process allowed these ridged electrodes to sustain fields of up to 60 MV/m . (This is lower than the typically achieved level in the LES,²⁰ possibly due to the altered shape of the electrodes.) Applied voltages were limited by this maximum sustainable field, as well as the maximum 1 mA obtainable current.

A specialized electric setup, shown in Fig. 1, was used to search for the subcritical fluctuations predicted by the MDDF model. The signal output is capacitively coupled to the high-voltage line, followed by input protection and low-noise amplifiers, before being sampled by an oscilloscope, with $1 \text{ }\mu\text{A}$ of dark current corresponding to approximately 1.75 mV measured by the oscilloscope. The system was designed for a broad bandwidth ($> 1.5 \text{ GHz}$) and low noise, to account for as large a range of possible signals as possible, since the characteristics of the signal were not known *a priori*. A power supply filter was

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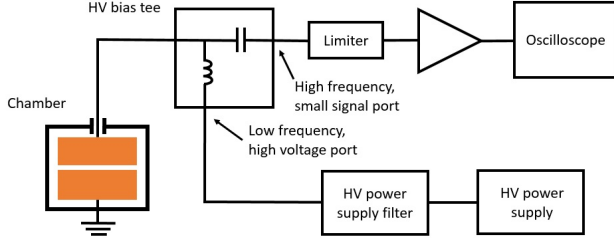


FIG. 1. Block diagram of the setup used to measure dark current fluctuations.

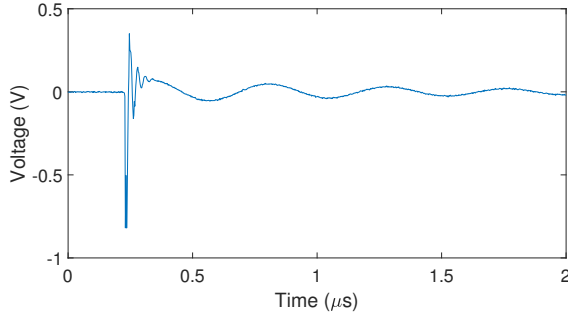


FIG. 2. Example oscilloscope trace of a subcritical (i.e., pre-breakdown) current spike, recorded at a high sample rate of 500 MHz.

added to attenuate any residual electrical noise from the switching behavior of the high-voltage power supply.

In both configurations, measurements of the dark current as a function of time showed pronounced brief current spikes occurring at varying intervals. These unique subcritical prebreakdown events, see, e.g., Fig. 2, consist of a fast transient of ~ 50 ns followed by a decaying sinusoid at ~ 2.5 MHz. The latter is thought to be the resonant response of the setup to the initial current spike. Measurements are performed via capacitive coupling, resulting in a high-pass response which removes the DC component of the dark current. This precludes the differentiation among transient spikes and step changes in the dark current. For obtaining a statistically significant amount of events, a low sampling rate of 5 MHz was used, thus optimizing oscilloscope memory usage. Although this rendered the initial spike too fast to resolve, the 2.5 MHz ringing remained visible, thus allowing to determine the time of occurrence per spike and their ensuing time distribution, which is the focus of this study.

Five series of measurements, as listed in Table I, were performed. Over the course of each series, the applied voltage was varied in steps. An algorithm was applied to assign a time of occurrence to each spike, avoiding double counting due to the ringing. Using this algorithm the distribution of time intervals between the spikes, for each voltage in each measurement series, was calculated. The CDF and frequency histogram of these distributions,

TABLE I. List of measurement series.

Series	Electrode configuration	Symbol in Fig. 5	β
A	crossed	circles	3.2
B	crossed	diamonds	1.8
C	parallel	triangles	1.7
D	parallel	squares	0.5
E	parallel	x	0.6

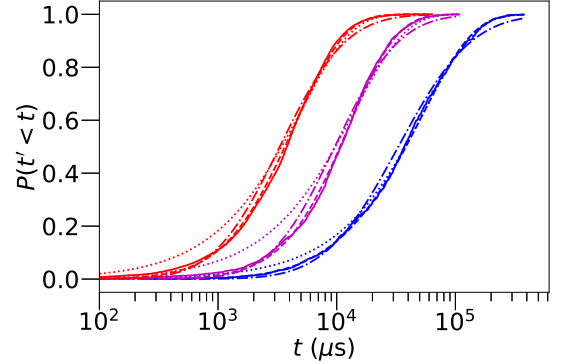


FIG. 3. CDF of the time interval between current spikes for surface fields of 31.3, 35.4, and 39.6 MV/m (from right to left), from series D. For each field, the CDF calculated from experiment is drawn as a solid line. The other lines represent best fits to hypoexponential (dashed), exponential (dotted), and log-normal (dash-dotted) distributions.

for a few example voltages in one series, are shown in Figs. 3 and 4, respectively.

Each measured distribution was characterized by an average rate of occurrence of spikes r_s . This rate as a function of the surface electric field E , for each measurement series, is shown in Fig. 5. The rates in Fig. 5 are given in units of spikes per second per slip plane, under the assumption that an active slip plane can develop in an area of approximately $\Delta\rho^{-2} = 10^{-4} \text{ mm}^2$.¹³

As seen in Fig. 5, spikes were observed at fields starting from a threshold field E_{th} , varying between 21 and 29 MV/m among series, and r_s was found to be an increasing function of $E - E_{\text{th}}$. However, as progressing series of measurements were made, this dependence became milder, and the overall values of r_s decreased. This decrease over time is in line with the proposal that the current spikes are connected to the breakdown rate, since high-voltage electrodes are known to undergo a conditioning process, in which exposure to strong electromagnetic fields over time makes them less susceptible to subsequent breakdown.^{17,19,20} The electric field and vacuum were retained between series, so that the electrodes were continuously being conditioned between them. This long-term conditioning mechanism does not preclude short-term deviations, as observed between series D and E.

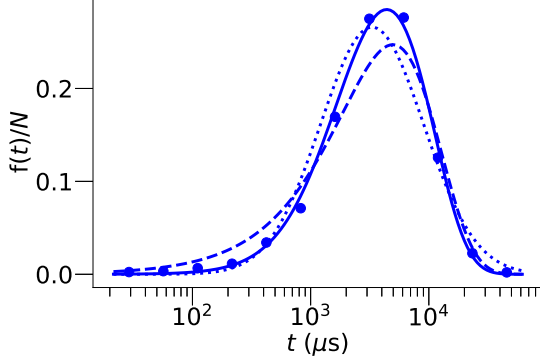


FIG. 4. Histogram of the time interval between current spikes for a surface field of 39.6 MV/m, from series D. The time intervals are divided logarithmically into thirteen bins, and the frequency axis is normalized by the total number of samples. The circles represent the experimental results (with error bars that are smaller than the size of the symbols), and the lines represent best fits to hypoexponential (solid), exponential (dashed), and log-normal (dotted) distributions. The logarithms of the likelihood ratios are $\ln(L_h/L_e) = 227$ for the hypoexponential and exponential fits, and $\ln(L_h/L_l) = 466$ for the hypoexponential and log-normal fits.

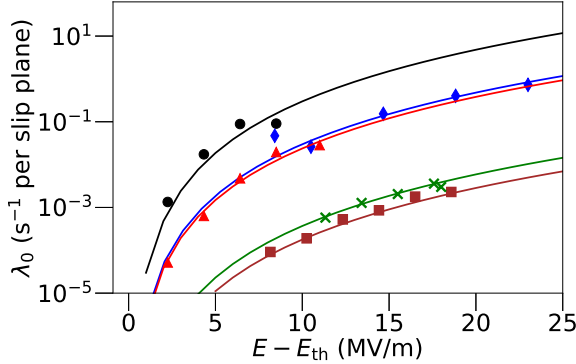


FIG. 5. Rate of current spikes as a function of the surface electric field, experimental (markers) and theoretical (lines), for the measurement series listed in Table I. The theoretical lines differ only in the parameter β , as detailed in the table.

Indeed, short-term increases of the breakdown rate as a function of the electric field are regularly observed in the conditioning process.

It was proposed in Ref. 13 that fluctuations of the mobile dislocation density within slip planes of a metal, subjected to a strong electric field, can be described as a birth-death Markov process. In this process the density n , representing the number of mobile dislocations in one slip plane, can change by ± 1 , normally fluctuating around $n_* = 0$, and ranging from 0 to a critical density n_c . When n reaches n_c a critical transition occurs,

leading to a runaway increase in n and, subsequently, to possible breakdown nucleation.^{12,13}

The corresponding birth and death rates of the Markov process are

$$\lambda_n = \frac{25\kappa C_t c}{G^2 b \Delta \rho} \sigma^2 e^{-\frac{E_a - \Omega \sigma}{k_B T}}; \quad \mu_n = \frac{50\xi C_t c}{G} \sigma n. \quad (1)$$

Here σ is the stress in the slip plane, given by $\sigma = \epsilon_0(\beta E)^2/2 + ZGb\Delta\rho n$, T is the temperature, and E is the electric field. The rest of the constants are physical values obtained by microscopy, *a priori* physical considerations, and fitting the MDDF theory to measured breakdown rates. It was shown that the variation among experimental structures of the breakdown rates, as a function of the electric field, can be explained as a difference in only one parameter, β .¹³

The breakdown rate is clearly not equal to r_s , since the latter is the rate of a prebreakdown phenomenon. To estimate r_s we consider that, prior to breakdown, each active slip-plane system is in a metastable state, in which the creation and arrest of mobile dislocations occur regularly. An increase in the density of mobile dislocations, in cells on the surface of the cathode, can be expected to modify the surface structure. This, in turn, may affect the dark current between the electrodes. We therefore consider the possibility that the spikes of dark current can be described in terms of the same kinetic model as that of the breakdown rate.

To analyze this hypothesis quantitatively, we consider an increase in mobile dislocations, i.e., a transition from any state n to the state $n+1$. The rate of such a transition is given by $\pi_n \lambda_n$, where π_n is the value of the quasistatic distribution function (see Refs. 12 and 13) at n . In the range of interest for experimental setups, $\lambda_0 \gg \lambda_{n>0}$, and also $\pi_0 \approx 1 \gg \pi_{n>0}$. The rate at which birth events occur is, therefore, approximately equal to λ_0 . In addition, since in the experimental range of interest we have $\mu_{n+1} \gg \lambda_n$ for every n (and in particular $\mu_1 \gg \lambda_0$), practically every birth event will be closely followed by a death event. λ_0 can be thought of, then, as the rate at which dark current spikes, each representing a birth-and-death pair of events, will occur. Therefore, λ_0 can be used to approximate r_s .

From Eq. (1), we find that

$$\lambda_0 = \frac{25\kappa C_t c}{G^2 b \Delta \rho} \sigma_0^2 \exp\left(-\frac{E_a - \Omega \sigma_0}{k_B T}\right). \quad (2)$$

Here, we might expect $\sigma_0 = \sigma(n=0) = \epsilon_0(\beta E)^2/2$. However, we saw in the experiment that current fluctuations occur only above E_{th} . This observation leads us to redefine the effective stress for mobile dislocation nucleation as

$$\sigma_0 = \frac{1}{2} \epsilon_0 [\beta (E - E_{th})]^2. \quad (3)$$

Such a threshold field might be due to preexisting stresses within the sample.

With this modified σ_0 , the values of λ_0 predicted by Eq. (2) can be directly compared to the measured dark current spike rates r_s . Since the dependencies of the spike rates on the electric field varied among measurement series, the parameter β is fitted individually for each series, while the values of all other constants remain as in Ref. 13. This is in line with the fitting procedure used previously to compare r_c to measured breakdown rates, where different values of β (and no other parameter) differentiated among experimental structures, thus accounting for both geometrical differences and different levels of conditioning of the electrodes.^{12,13} It is also in line with the fact that β decreases over time, due to conditioning. As seen in Fig. 5, the theoretical values of λ_0 as a function of E provide a good fit to the experimental results, with no need to further adjust the MDFF theory or any numerical values. This supports the hypothesis that the dark current spikes are, indeed, a manifestation of the fluctuations in the mobile dislocation density population as described in the MDFF model.

Having established that λ_0 gives a good approximation of r_s , we further assume that each observable variation of the dark current is directly related to a mobile dislocation reaching the surface of the sample. Under this assumption, the full distribution of times between current spikes can be evaluated using the MDFF model. We recall that in the MDFF model, mobile dislocations are free to move within subgrain dislocation cells.¹³ Here, we need to consider the kinetics of mobile dislocations in cells adjacent to the surface, and the probability of their ejection to the surface. In general, mobile dislocations can become sessile due to either reaching the surface or collision with barriers.^{12,13} However, we expect cells adjacent to the surface to be relatively denuded of the sessile dislocations which serve as barriers, due to their strong elastic interaction with the surface.²¹

The process of a mobile dislocation reaching the surface involves, therefore, two steps: (i) The creation of a mobile dislocation, and (ii) the ejection of the dislocation to the surface. These are represented in the MDFF model as transitions from state n to $n + 1$, followed by a subsequent transition from state $n + 1$ to n . Assuming that $\mu_1 \gg \lambda_1$, as is the case in the experimental range of interest, the time between between spikes can be approximated by considering the random variable $T = T_{0 \rightarrow 1} + T_{1 \rightarrow 0}$. Here, $T_{0 \rightarrow 1}$ and $T_{1 \rightarrow 0}$ are random variables describing the transition time from state $n = 0$ to $n = 1$ and back, respectively. Assuming, for simplicity, that each transition individually is a poissonian process, we find that T is a two-parameter hypoexponentially distributed variable, with the rate parameters λ_0 and μ_1 .²²

For each voltage in each measurement series, the observed distribution of the time intervals between fluctuations was fitted to an exponential, two-parameter hypoexponential, and log-normal distribution (using MATLAB's `mle` function). The two-parameter hypoexponential fits yielded better results in the majority of the cases, and particularly for the higher voltages, in which

there were more samples per voltage. The sum of the logarithms of the likelihood ratios²³ of all the fits was $\Sigma \ln(L_h/L_e) = 5.16 \times 10^5$ for the hypoexponential and exponential fits, and $\Sigma \ln(L_h/L_l) = 5.73 \times 10^3$ for the hypoexponential and log-normal fits. (For this calculation, the data measured at 2125 V in series B was excluded, as fitting it to any of the three distributions yielded poor results.) The quality of fit is demonstrated in Figs. 3 and 4 for three out of the twenty-five voltage steps in the measurement series.

The MDFF model provides a qualitative and quantitative explanation of observed breakdown rates in high-voltage applications. The purpose of the experiment described in this manuscript was to investigate the suggestion that the fluctuations of the mobile dislocation density, which are an underlying feature of the model, can be observed before breakdown itself occurs.^{12,13} Indeed, dark current spikes were discovered, and their rates match the theoretical state transition rate $\lambda_0(E)$ remarkably well, with no need to adjust the previously derived numerical parameters of the model.

The fit of the time interval distribution to a hypoexponential distribution further supports the proposal that the current spikes are an inherent feature of the dynamics described by the MDFF model. The fitted value of λ_0 can be compared to r_s . However, since $\mu_1 \gg \lambda_0$, the confidence interval for the fit of μ_1 is large, making it impractical to derive values of μ_1 as a function of the electric field.

The experiment described here was performed on ridged OFHC copper electrodes. By modifying the electrode material production process and geometry, it might be possible to vary the amount of spikes prior to breakdown. This would enable us to study the dependence of prebreakdown fluctuations on the electrode structure. A configuration in which the spike rate increases significantly would allow more data samples to be collected per measurement, allowing the estimation of μ_1 , and maybe even the exploration of λ_n for $n > 0$. In addition, the current experimental setup allows the identification of strong temporal variation in the current, but not its specific shape. Thus, the current work concentrated on reproducing the distribution of times between current spikes. Future experiments can investigate the form of the emitted current within spikes, possibly providing constraints on the specific current emission mechanism.

The observed correlation between current spikes and breakdown can be utilized to reduce the breakdown rate in high voltage setups, and improve operational procedures in such systems, by using rates of observed dark current spikes to determine the level of conditioning of electrodes, or by monitoring spikes as observed *in situ* prior to breakdown.

See supplementary material for further details of the experimental procedure and spike identification algorithm.

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The data that support the findings of this study are available from the corresponding author upon reasonable request.

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